NEW TEST SET – OPEN SET EVALUATION AGAINST PREVIOUS TRAIN DATA

System Overview

This system performs open-set biometric verification of cattle using two visual modalities: face and muzzle. The goal is not to classify a sample into a closed set of known identities, but to determine whether an input sample belongs to any enrolled identity in the gallery.

The pipeline is explicitly designed for **unknown-only open-set evaluation**, where all test identities are unseen during training.

Input Specification

Input Modalities

Each evaluation sample consists of two inputs:

1. **Face Image**
   A cropped RGB image of the cattle's face.
2. **Muzzle Image**
   A cropped RGB image of the same cattle's muzzle.

Both images are resized to **224 × 224**, normalized using ImageNet statistics, and passed independently through modality-specific embedding networks.

Model Architecture

Two separate embedding models are used:

- **Face Embedding Model**
  A ResNet-50 backbone followed by a 512-dimensional projection head.
- **Muzzle Embedding Model**
  An identical architecture trained independently on muzzle images.

Each model outputs an L2-normalized embedding vector. Cosine similarity is used as the distance metric.

Training Data

The training dataset consists of cattle identities with paired face and muzzle images.

From this dataset:

- Face embeddings were extracted to form the **face gallery**
- Muzzle embeddings were extracted to form the **muzzle gallery**

Each gallery contains one representative embedding per enrolled identity. These galleries serve as the reference set for all evaluations.

The trained face and muzzle models were saved after training and **not updated during testing**.

NEW TEST SET – OPEN SET EVALUATION AGAINST PREVIOUS TRAIN DATA

Test Data

Original Open-Set Test

The initial open-set test dataset contains only unknown cattle identities that are not present in the training gallery.

This dataset was used for image-wise open-set evaluation.

New Structured Open-Set Test

A second open-set test dataset was **orchestrated and structured** to enable both image-wise and cow-wise evaluation.

Directory structure:

```
cattledatarefined/
└── all_data_part_1/
    ├── C1/
    │   ├── face/
    │   └── muzzle/
    ├── C2/
    │   ├── face/
    │   └── muzzle/
```

Each cattle identity contains all face and muzzle images grouped under a single folder. None of the cattle identities overlap with the training set.

Inference and Decision Process (Corrected)

Each test sample consists of a face image and a muzzle image of the same cattle.
The system applies a two-stage gated verification pipeline, where both modalities are thresholded.

Step 1: Face-Based Search (1:N)

The face image is passed through the face embedding model to obtain a normalized embedding vector.
This embedding is compared against all entries in the face gallery using cosine similarity.

From this comparison:

- **The maximum similarity score and its corresponding gallery identity are selected.**

- **This score is evaluated against a face similarity threshold.**

Decision at this stage:

- **If the face similarity score is below the face threshold, the sample is rejected.**
- **If the face similarity score meets or exceeds the threshold, the predicted identity is tentatively selected and the system proceeds to muzzle verification.**

Step 2: Muzzle Verification (1:1)

The muzzle image is independently passed through the muzzle embedding model.
The resulting embedding is compared only against the muzzle gallery entry corresponding to the face-predicted identity.

This comparison produces a muzzle similarity score, which is evaluated against a muzzle similarity threshold.

Decision at this stage:

- **If the muzzle similarity score is below the muzzle threshold, the sample is rejected.**
- **If the muzzle similarity score meets or exceeds the threshold, the sample is accepted.**

This two-stage gating prevents false acceptance caused by a single modality.

Fusion Mechanism

Let

- S_{face} be the maximum cosine similarity score obtained from face-based search (1:N)
- S_{muzzle} be the cosine similarity score obtained from muzzle verification (1:1)

The fused score is computed as:

$$S_{fused} = 0.6 \cdot S_{face} + 0.4 \cdot S_{muzzle}$$

This weighting reflects the higher discriminative stability of the face modality, while still enforcing confirmation from the muzzle modality.

Decision Rule

The final decision is made by comparing the fused score against a fusion threshold T_{fusion}:

$$\text{Accept} \iff S_{fused} \geq T_{fusion}$$

If the fused score is below the threshold, the sample is rejected.

NEW TEST SET – OPEN SET EVALUATION AGAINST PREVIOUS TRAIN DATA

Evaluation Protocol

All evaluations follow an **unknown-only open-set protocol**:

- The test set contains no enrolled identities
- True positives do not exist by definition
- The most critical metric is False Accept Rate (FAR)

Metrics Defined

- **False Positive (FP):** An unknown cattle sample accepted as known
- **False Accept Rate (FAR):**
- **True Positives (TP):** 0 by definition
- **True Accept Rate (TAR):** 0 by definition

Results: Image-Wise Open-Set Evaluation

Metric	Value
Total Test Samples	916
True Positives (TP)	0
False Positives (FP)	0
True Accept Rate (TAR)	0.0000
False Accept Rate (FAR)	0.0000

Results: Cow-Wise Open-Set Evaluation

For cow-wise evaluation, all face images of a cattle identity were evaluated first and their similarity scores averaged. The dominant predicted identity from the face modality was then used for muzzle verification, where all muzzle similarity scores were also averaged.

A cattle identity was accepted only if both the **average face score** and **average muzzle score** exceeded their respective thresholds.

Metric	Value
Total Cattle Evaluated	200
Accepted Cattle	0

NEW TEST SET – OPEN SET EVALUATION AGAINST PREVIOUS TRAIN DATA

Metric	Value
Rejected Cattle	200
Cow-Level FAR	0.0000

Interpretation

The absence of false accepts in both image-wise and cow-wise evaluations demonstrates that the system reliably rejects unseen cattle identities.

The results confirm:

- strong separation between enrolled and unseen identities
- effective threshold calibration
- robustness of the two-factor face–muzzle verification pipeline

This behavior is critical for real-world biometric deployment, where rejecting unknown identities is often more important than aggressive acceptance.